

Alibek T. Kaliyev

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Education

University of Texas at Austin

Master of Science, Computer Science

Part-time studies with a specialization in Machine Learning

GPA: 4.00/4.00

Coursework: Deep Learning — Advances in Deep Learning — Reinforcement Learning — Parallel Systems

Austin, TX

Aug 2024 - Dec 2027 (Expected)

Lehigh University

Bachelor of Science, Computer Science and Business

Minor: Cognitive Science

GPA: 3.78/4.00

Computer Science Coursework: Computer Vision — Fundamentals of Machine Learning — Biomedical Image Computing — Operating Systems Design — Algorithms & Data Structures — Database Systems, Algorithms & Applications — Design and Analysis of Algorithms — Blockchain Systems, Algorithms and Applications — Software Engineering — Computer Organization & Architecture — Systems Software — Programming Languages — Fast Machine Learning.

Mathematics Coursework: Calculus (I, II) — Statistical Methods — Linear Methods — Foundations of Discrete Structures & Algorithms.

Bethlehem, PA

June 2020 - May 2024

Publications

A.T. Kaliyev, R. Forelli, P. Sales, S. Qin, Y. Guo, S.O. Memik, M.W. Mahoney, A. Gholami, R.K. Vasudevan, S. Jesse, N. Tran, P. Harris, M. Takáč, J.C. Agar. Rapid Fitting of Band-Excitation Piezoresponse Force Microscopy Using Physics Constrained Unsupervised Neural Networks. **AI for Accelerated Materials Design NeurIPS 2023 Workshop**, Dec 2023, New Orleans, LA

S. Qin, Y. Guo, **A.T. Kaliyev**, J.C. Agar, 2022. Why it is Unfortunate that Linear Machine Learning Models “Work” so well in Electromechanical Switching of Ferroelectric Thin Films. *Advanced Materials*. 2202814.

Presentations

Gulf Coast Undergraduate Research Symposium 2022 (Rice University). “Rapid Fitting of Band-Excitation Piezoresponse Force Microscopy and Piezoelectric Hysteresis Loops Using Physics Constrained Unsupervised Neural Networks.” Oct 2022, Houston, TX.

Fast Machine Learning for Science Workshop 2022 (SMU). “Rapid Fitting of Band-Excitation Piezoresponse Force Microscopy and Piezoelectric Hysteresis Loops Using Physics Constrained Unsupervised Neural Networks.” Oct 2022, Dallas, TX.

MSE Undergraduate Research Symposium (Lehigh University). “Rapid Fitting of Band-Excitation Piezoresponse Force Microscopy and Piezoelectric Hysteresis Loops Using Physics Constrained Unsupervised Neural Networks.” Sep 2021, Bethlehem, PA.

Drexel AI Research Conference. “Accelerated Fitting of Band-Excitation Piezoresponse Force Microscopy.” May 2021, Remote.

David and Lorraine Freed Undergraduate Research Symposium (Lehigh University). “Accelerated Fitting of Band-Excitation Piezoresponse Force Microscopy.” Apr 2021, Bethlehem, PA.

Gulf Coast Undergraduate Research Symposium 2020 (Rice University). “Pre-trained Deep-Learning Models for Rapid Analysis of Piezoelectric Hysteresis Loops.” Oct 2020, Remote.

Center for Nanophase Materials Sciences User Meeting (ORNL). “Pre-trained Deep-Learning Models for Rapid Analysis of Piezoelectric Hysteresis Loops.” Aug 2020, Remote.

Research Experience	Multifunctional Materials and Machine Learning Group	Bethlehem, PA
	<i>Undergraduate Deep Learning Researcher</i> ◦ Accelerated extraction of mechanical properties of quantum materials by 3.5 times by building a deep neural network with convolutional layers in the form of an autoencoder. ◦ Optimized a neural network's training and inference processes in PyTorch by reducing the model size by a factor of 3,000, and performing a quantized-aware training using hls4ml. ◦ Visualized and preprocessed 1.3 million data samples from HDF5 files by leveraging Matplotlib, Seaborn, Scikit-learn, and NumPy. ◦ Reduced the number of bits needed to train the neural network by 54% while keeping the same MSE by performing quantization-aware training using AutoQKeras. ◦ Prepared the deployment of the model on Xilinx K60 FPGA using HLS4ML library for fast and real-time inference; benchmarks demonstrate the latency of 40 microseconds/fit. <i>PI: Dr. Joshua C. Agar</i>	June 2020 - Dec 2023
	Brain Imaging & Computation Lab	Bethlehem, PA
	<i>Undergraduate Research Assistant</i> ◦ Developed Graph Attention Neural Networks for depression severity prediction in patients achieving R^2 value of 0.91 between predicted and true values of Beck Depression Inventory (BDI) and Spielberger Trait Anxiety Inventory (TAI) <i>PI: Dr. Yu Zhang</i>	Aug 2022 - Mar 2023
Industry Experience	Amazon Web Services	New York City, NY
	<i>Software Development Engineer</i> ◦ Leading development of a new generative AI product as one of the first engineers on a 6-person team, building the entire technical foundation from scratch ◦ Architected and implemented complete infrastructure including VPC networking, DNS, Elastic Load Balancers, CI/CD pipelines, security frameworks, and testing infrastructure for hosting on ECS Fargate ◦ Established technical leadership role by becoming the go-to person for architecture decisions and implementation guidance as the product matures toward launch ◦ Led end-to-end development of active user metrics for Amazon Q's Business Analytics Dashboard, delivering critical analytics capabilities to over 500,000 monthly enterprise customers ◦ Discovered and resolved critical upstream microservice issue that was silently losing 100,000+ customer requests, coordinating cross-team resolution and data backfill strategy ◦ Engineered performance optimization solutions that reduced Lambda execution latency by 75%, significantly improving system responsiveness and reducing operational costs	June 2024 - Present
	Capstone Project for Merck & Co.: Machine Assisted Contextualization	Bethlehem, PA
	<i>Machine Learning Engineer/Researcher</i> ◦ Fast-forwarded team's development of machine learning pipeline to classify manufacturing labels using Scikit-learn, Pandas and NumPy ◦ Improved an accuracy of predictions from 47% to 92% on the test set by utilizing feature engineering methods, which optimized the labeling process at Merck by 10x ◦ Further researched and developed state-of-the-art semi-supervised GAN-BERT model to improve the accuracy of labeling algorithm to 96% on the test set ◦ Integrated the human-in-the-loop component of the project by deploying the model on AWS using FastAPI and Nginx to enable the retraining process <i>Advisor/sponsor: Vincent R. Capodanno</i>	Jan 2023 - Dec 2023
	Amazon Web Services (Edge ML Services Team)	New York City, NY
	<i>Software Development Engineer Intern</i> ◦ Designed and implemented an automated CloudFormation stacks updater in the CI/CD production pipeline, which resulted in the team's deployment 15 times faster ◦ Shipped a production-level and ready code 3 times faster than expected, saving the team's time to deliver updates for customers. Utilized AWS CDK, Step Functions, Lambda, VPC and Custom Resources ◦ Communicated effectively with senior engineers across 3 teams to discuss and implement the service into their AWS back-end. Worked on unblocking pipelines and handling failure behaviors ◦ Tracked and monitored behavior of the service by implementing CloudWatch alarms and ticketing	May 2023 - Aug 2023

actions

GrainBound, LLC

Machine Learning Technical Associate

Bethlehem, PA

Feb 2022 - May 2023

- Developed and applied machine learning techniques and algorithms to provide consulting services for a large international chemicals manufacturer
- Improved the manufacturing efficiency by 40% by providing insights on dataset with 50,000 rows using Pandas, and predicting the manufacturing settings for each material using Tensorflow and Scikit-learn

Teaching Experience **Lehigh University, Computer Science & Engineering Department**

Course Assistant & Grader

Computer Vision (CSE 398/498)

Spring 2024

Database Systems & Applications (CSE 241)

Fall 2023

Data Structures & Algorithms (CSE 017)

Fall 2022 & Spring 2023

Introduction to Programming (CSE 007)

Spring & Fall 2021

Honors & Awards

Trustees Scholarship (given to 1% of applicant pool), \$30,000/year

2020-2024

Lehigh Grant, \$35,000/year

2020-2024

Beta Gamma Sigma Business Honors Society (top 10% undergrad)

2023

Lehigh University Dean's List

Fall 2020 - Spring 2023

Nano-Human Interfaces Presidential Fellowship, \$8,000

2022

STEM-SI Fellowship, \$5,500

2021

Most Novel Research Award at Drexel AI Conference

2021

Facebook ABCS Fellowship

2021

Data for Impact Fellowship

2020

Community
Involvement

Secretary, Lehigh Central Asian Students Association

Mar 2023 - May 2024

Vice President, Lehigh Comp. Sci. & Business Association

Apr 2022 - Apr 2023

Technical Development Chair, Lehigh Comp. Sci. & Business Association

Apr 2021 - Apr 2022

Skills

Programming Languages: Python, Java, TypeScript, C/C++, SQL, HTML/CSS

ML/DS Libraries: NumPy, SciPy, PyTorch, TensorFlow, Pandas, Matplotlib

Software Engineering: AWS, React, Docker, Git, Google Cloud, Heroku

Languages: English, Russian, Kazakh